Syllabus ESM 262 - Spring 2026

A bit about me

Figure 1: Growing skills

Course description

ESM 262 is an introduction to computing for environmental applications. The course provides practical training in software design best practices.

- programming language concepts
- loops and repetition
- data types and structures
- modular program design;
- reproducibility and version control;
- testing, documentation, and workflow;

The course features **R** for programming, **Git** for version control, **Quarto** for workflow and editing, and **GitHub** for collaboration and publishing, but many concepts would be applicable in other software design tools.

Class will include a mix of lectures and hands-on examples, using students' own computers.

Course where and when

T/TH 8:00 AM - 9:15 AM (Bren Hall 1414)

Teaching Team

Instructor: Naomi Tague (www.tagueteamlab.org)

- **Office hours:** email me
- **email** (tague@ucsb.edu)
- **git user id** naomitague

Teaching assistant: Ojas Sarup

- **Office hours:** TBD
- **email** (ojassarup@ucsb.edu)
- **git user id** osarup

Learning objectives

- practice partner coding and reproducible workflows
- 3 *BIG* concepts in programming are *modularity*, *data structure*, *repetition* - we will learn skills related to all three of these
- learn and practice some coding/programming best practices (that will make your data science life easier - whatever you do!)

  - documentation
  - testing
  - reproducibity

- learn some new Rskills that are helpful for a wide variety of data science applications

Computing set up

- R, R-studio
- Please bring your laptop to class

Course materials and how we will work

All lecture materials will be available on this course website as well as assignments. (But you will submit assignments on Canvas)

Prior to many classes, I will ask you to study an Quarto document (and often an example R-functions) before class Then at the beginning of class, I will **briefly** go over the document and we will discuss any questions. Occasionally I spend 10-15 minutes lecturing. We will spend most of the class working on practical applications of what was in the document. This way we use the class time as a lab - where you can get guidance.

this mean its essential for you to do the review before class

Why structure it this way? You learn to code by doing - not so much by watching others do!

Topics

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

#	Date	Type	Topic
1	Tue, Apr 14	Regular	Course overview and setup
2	Thu, Apr 16	Regular	Quarto
3	Tue, Apr 21	Regular	Git and GitHub
4	Thu, Apr 23	Regular	Git and conflicts
5	Tue, Apr 28	Regular	Data structures - Strings and Datas
6	Thu, Apr 30	Regular	Data Types and Structures, Lists
7	Tue, May 5	Regular	APIs and servers
8	Thu, May 7	Regular	Loops
9	Mon, May 11	Evening	Functions!
10	Tue, May 12	Regular	Functions 2
11	Thu, May 14	Regular	Repeating
12	Mon, May 18	Evening	Project Design, Modularity and Workflow - Practicing
13	Tue, May 19	Regular	Flow Control
14	Thu, May 21	Regular	Modular design
15	Tue, May 26	Regular	Working with the shell
16	Thu, May 28	Regular	Testing and debugging
17	Tue, Jun 2	Regular	Documentation
18	Thu, Jun 4	Regular	Where does AI fit

Assignments

There will be 8 assignments (more or less one for each week). You will usually have time to work on the assignment in class and most will be in groups.

Assignment materials and dates

Week	Name	Assigned	Due date	Group (G)
1	Quarto Intro	2026-04-16	2026-04-21	G
2	Git	2025-04-21	2025-04-28	I
3	Using Git with your function	2025-02-25	2025-03-03	I
4	Data Structure	2025-03-4	2025-03-10	I
5	Functions and testing	2025-3-11	2025-3-18	G

Assignment expectations and grading

- All assignments submitted on Canvas

- Late Assignments 10% each day. Try to submit on time as we build on the concepts and materials covered in each assignment
- Style counts; So make sure your follow good programming practices including adding documentation and using informative variable names
- You can use AI (ChatGPT etc) to help you learn syntax, but write the code yourself; this is how you learn
- You can resubmit your assignment for additional feedback and we will upgrade your grade. Please do so with in one week of getting the assignment back